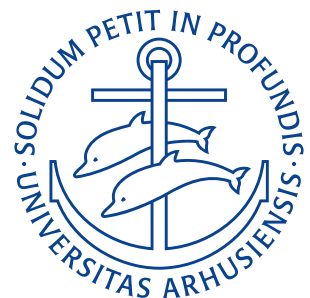


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Exercises

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Course: Fluid Mechanics — 290211U004
Date: January 21, 2026



Contents

1	Lecture 25+: Exam 2024	2
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Lecture 25+: Exam 2024

January, 21 2026

Problem 1

For the static system shown in Figure 1, determine the gage pressure in kPa at point a . Liquid A has an SG = 1,2 and liquid B has an SG = 0,75. The liquid surrounding point a is water ($\rho = 1000 \text{ kg/m}^3$) and the tank on the left is open to atmosphere

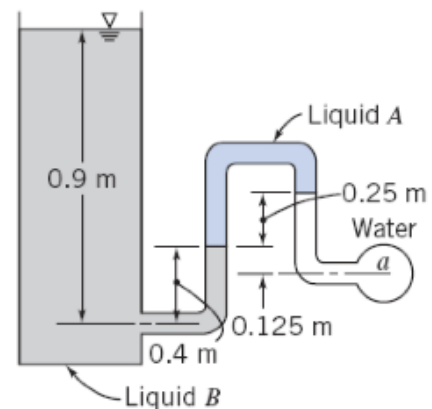


Figure 1

Assumptions

- Hydrostatic equilibrium
- All fluids are at rest and will not mix and will not exert any capillary effects on each other
- All liquids have constant densities
- The left tank is open to the atmosphere, i.e. $p_b = 0$ gauge at the surface

Derivation

The heights of the different liquid-interfaces is understood as:

- Free surface of B is at $z = 0,9 \text{ m}$
- AB interface is at $z = 0,4 \text{ m}$
- AW interface is at $z = 0,4 \text{ m} + 0,25 \text{ m} = 0,65 \text{ m}$
- Point a is at $z = 0,125 \text{ m}$

We perform a “hydrostatic walk” from the free surface to point a . From the free surface to the AB interface we move vertically downwards a distance if $B_z - AB = 0,9 \text{ m} - 0,4 \text{ m} = 0,5 \text{ m}$. This corresponds to a pressure change of:

$$p_{AB} = \rho_B g \cdot (B_z - AB) = 0,75 \cdot 1000 \text{ kg/m}^3 \cdot 9,82 \text{ m/s}^2 \cdot 0,5 \text{ m} = 3,68 \text{ kPa}.$$

Then we “move” a distance of $AB - AW = 0,4 \text{ m} - 0,65 \text{ m} = -0,25 \text{ m}$ which corresponds to a pressure

$$p_{AW} = p_{AB} - \rho_A \cdot g(AB - AW) = 3,68 \text{ kPa} - 1,2 \cdot 1000 \text{ kg/m}^3 \cdot 9,82 \text{ m/s}^2 \cdot 0,25 \text{ m} = 0,737 \text{ kPa}.$$

Then we move a distance of $AW - a_z = 0,65 \text{ m} - 0,125 \text{ m} = 0,525 \text{ m}$ which corresponds to a pressure

$$p_a = p_{AW} + \rho_w g(AW - a_z) = 0,737 \text{ kPa} + 1000 \text{ kg/m}^3 \cdot 9,82 \text{ m/s}^2 \cdot 0,525 \text{ m} = 5,89 \text{ kPa}.$$

Hence, the gage pressure at a is about $p_a = 5,89 \text{ kPa(gage)}$.

$p_a = 5,89 \text{ kPa}$

RESULT

Problem 2

Water ($\rho = 1000 \text{ kg/m}^3$) is flowing steadily through the 180° elbow shown in Figure 2. At the inlet to the elbow the gage pressure is 103 kPa. The water discharges into atmospheric pressure. Assume that the flow properties are uniform over the inlet and outlet areas; $A_1 = 2500 \text{ mm}^2$, $A_2 = 650 \text{ mm}^2$, and $V_1 = 3 \frac{\text{m}}{\text{s}}$ are given. Find the horizontal component of force required to hold the elbow in place.

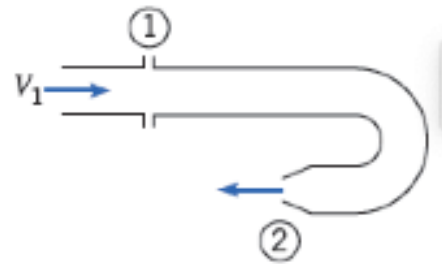


Figure 2

Assumptions

- Steady, uniform flow at each section
- Incompressible fluid
- Negligible wall shear

Derivation

Due to mass conservation, the mass flow rate must be constant, $Q = \text{const}$. I.e.

$$Q = A_1 V_1 = 2500 \text{ mm}^2 \cdot 3 \frac{\text{m}}{\text{s}} = 0,0075 \text{ m}^3/\text{s} \implies V_2 = \frac{Q}{A_2} = \frac{0,0075 \text{ m}^3/\text{s}}{650 \text{ mm}^2} = 11,5 \frac{\text{m}}{\text{s}}.$$

As the outlet points left $V_{2x} = -11,5 \frac{\text{m}}{\text{s}}$. This means the mass flow is

$$\dot{m} = \rho Q = 1000 \text{ kg/m}^3 \cdot 0,0075 \text{ m}^3/\text{s} = 7,5 \frac{\text{kg}}{\text{s}}.$$

For steady flow, the linear momentum balance in x is:

$$\sum F_x = \int_{\text{CS}} u \rho \mathbf{V} \cdot d\mathbf{A} = \dot{m} (V_{2x} - V_{1x}) = 7,5 \frac{\text{kg}}{\text{s}} \cdot \left(-11,5 \frac{\text{m}}{\text{s}} - 3 \frac{\text{m}}{\text{s}} \right) = -109 \text{ N}.$$

The negative sign indicates, that the net external force on the fluid points to the left.

At the inlet the pressure acts normal into the elbow as the outward normal points to the left:

$$F_{p1,x} = p_1 A_1.$$

At the outlet the water discharges to the atmosphere so here the pressure $F_{p2,x} = 0$ as $p_2 = 0$. This means, that the full force balance on the elbow in the x -direction is

$$\sum F_x = p_1 A_1 + F_{\text{wall},x}.$$

The sum of forces in the x -direction was previously found to be $\sum F_x = -109 \text{ N}$, meaning

$$p_1 A_1 + F_{\text{wall},x} = -109 \text{ N} \implies F_{\text{wall},x} = -109 \text{ N} - p_1 A_1 = -109 \text{ N} - 103 \text{ kPa} \cdot 2500 \text{ mm}^2 = -0,367 \text{ kN}.$$

Hence, the elbow will be pushed into the wall with a force of $F_{\text{fluid}} = 367 \text{ N}$ and the wall must supply a force $F_{\text{wall},x} = -367 \text{ N}$

$F_{\text{wall},x} = -367 \text{ N}$

RESULT

Problem 3

Consider the velocity field $\mathbf{V} = A(x^4 - 6x^2y^2 + y^4)\hat{i} + A(4xy^3 - 4x^3y)\hat{j}$, where $A = 0,25 \text{ m}^3/\text{s}$. Show that the flow field is incompressible. Calculate the acceleration of a fluid particle at point $(x, y) = (2, 1)$.

Assumptions

- As no t components are present in the velocity field it is implied to be steady

Derivation

For an incompressible flow the continuity equation simplifies to

$$\nabla \cdot \mathbf{V} = 0 \implies \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

We compute the partial derivatives:

$$\begin{aligned} u &= A(x^4 - 6x^2y^2 + y^4) & \implies & \frac{\partial u}{\partial x} = A(4x^3 - 12xy^2) \\ v &= A(4xy^3 - 4x^3y) & \implies & \frac{\partial v}{\partial y} = A(12xy^2 - 4x^3). \end{aligned}$$

I.e.

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = A(4x^3 - 4x^3 + 12xy^2 - 12xy^2) = 0$$

meaning the flow field is incompressible everywhere.

For the acceleration we use that:

$$\mathbf{a} = (\mathbf{V} \cdot \nabla) \mathbf{V}$$

which, in Cartesian coordinates is

$$a_x = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}, \quad a_y = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y}.$$

We compute the two missing derivatives as:

$$\begin{aligned} \frac{\partial u}{\partial y} &= A(4y^3 - 12x^2y) \\ \frac{\partial v}{\partial x} &= A(4y^3 - 12x^2y). \end{aligned}$$

So the acceleration components are

$$\begin{aligned} a_x &= u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \\ &= A^2(x^4 - 6x^2y^2 + y^4)(4x^3 - 12xy^2) + A^2(4xy^3 - 4x^3y)(4y^3 - 12x^2y) \\ &= 4A^2x(x^2 + y^2)^3 \\ a_y &= u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \\ &= A^2(x^4 - 6x^2y^2 + y^4)(4y^3 - 12x^2y) + A^2(4xy^3 - 4x^3y)(12xy^2 - 4x^3) \\ &= 4A^2y(x^2 + y^2)^3. \end{aligned}$$

So,

$$a_x(2, 1) = 4 \cdot (0,25)^2 \cdot 2(4 + 1)^3 = 62.5$$

$$a_y(2, 1) = 4 \cdot (0,25)^2 \cdot 1 \cdot (4 + 1)^3 = 31,25.$$

So,

$$a(2, 1) = 62,5 \text{ m/s}^2 \hat{i} + 31,3 \text{ m/s}^2 \hat{j}.$$

$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$ $a(2, 1) = 62,5 \text{ m/s}^2 \hat{i} + 31,3 \text{ m/s}^2 \hat{j}.$
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RESULT

Problem 4

A model airplane has a span of 120 cm and a chord of 25 cm. The model flies in air at standard atmospheric conditions (i.e., $\rho = 1,2 \text{ kg/m}^3$, $\nu = 1,5 \cdot 10^{-5} \text{ m}^2/\text{s}$). The wing can be regarded as a flat plate so far as drag is concerned. At what speed will a transition to a turbulent boundary layer start? What is the total drag force on the wing just before the transition to turbulence happens?

Assumptions

- The wing is flat and has zero thickness
- Flow is parallel to the wing, meaning to lift effects are present
- The freestream is uniform incompressible ($M \ll 0,3$), and low-turbulence
- The surface of the wing is assumed smooth
- Transition to turbulent flow happens at $\text{Re}_{x,\text{cr}} = 5 \cdot 10^5$

Derivation

The Reynolds number for the flow over the flat plate (the wing) is

$$\text{Re}_x = \frac{Vx}{\nu}, \quad \text{Re}_{x,\text{cr}} \approx 5 \cdot 10^5.$$

Due to the length-parameter of the Reynolds number transition to turbulent flow will happen at the trailing edge first. Inserting the chord length as our characteristic length and solving for the critical speed we obtain:

$$V_{\text{cr}} = \frac{\text{Re}_{x,\text{cr}} \nu}{L} = \frac{5 \cdot 10^5 \cdot 1,5 \cdot 10^{-5} \text{ m}^2/\text{s}}{0,25 \text{ m}} = 30 \frac{\text{m}}{\text{s}}.$$

The drag coefficient for flow with free stream velocity V , over a flat plate of length L and width b is:

$$C_d = \frac{1,33}{\sqrt{\text{Re}_L}} = \frac{1,33}{\sqrt{5 \cdot 10^5}} = 1,88 \cdot 10^{-3}.$$

The drag force F_D is

$$F_D = \frac{1}{2} \rho V^2 A C_D.$$

The area which the friction applies over A is in this case both the bottom and top of the wing, hence:

$$A = 2 \cdot 120 \text{ cm} \cdot 25 \text{ cm} = 0,6 \text{ m}^2.$$

So

$$F_D = \frac{1}{2} \cdot 1,2 \text{ kg/m}^3 \cdot \left(30 \frac{\text{m}}{\text{s}}\right)^2 \cdot 0,6 \text{ m}^2 \cdot 1,88 \cdot 10^{-3} = 0,61 \text{ N}.$$

$V_{\text{cr}} = 30 \frac{\text{m}}{\text{s}}$ $F_D = 0,61 \text{ N}.$
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RESULT